

University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Mid Term Examination, Autumn 2024

Course Title: Digital Logic Design

Course Code: CSE 215

Marks: 20

Time: 1(one) hour

(Answer all questions)

Q.No.	Questions	Marks
1a	Explain the terms: Associative law and Distributive law. Also, find the 7's and 10's complement of $(874)_{10}$.	4
1b	Calculate the following unsigned binary numbers: $(+36)_{10} + (-45)_{10} - (+23)_{10}$ $(76)_{10} - (45)_{10} + (13)_{10}$	4
1c	Suppose we have 4 variables (a, b, x, w) . Identify the minterm or maxterm by giving an explanation from the given expressions. i. $a + x' + b + w$ ii. $ab + w + x$ iii. $wx'a'b$ iv. $a'bxq$	2
2a	Design the Boolean expression of $E = ((A.B)' \cdot (C.D)')'$ using NAND gates.	3
2b	Given the Boolean functions: $F1 = W.Y + W.Y.Z' + W.X.Z + W'.X.Y'$ $F2 = W.Y + W'.X.Z' + X'.Y.Z' + X.Y'.Z$ Prove that $F1 = F2$ using the truth table.	3
2c	Find the complement of $F = WX + YZ$ and prove that $(F.F') = 0$ and $(F + F') = 1$.	4